

H1 — ML + Embedded Track

Member: _____

Your role

You run the fatembed arm (seed 1). You own the host golden verification and the on-chip profiling.

Weekly tasks

Week 1 — Golden tests

- Task: Set up the local Python env. Build the portable C runtime on host. Run the host golden tests (runtime/host_verify, tests/).
- Deliverable: The golden test results with the max abs diff.
- Verify: The diff is 1e-5 or below, as published.
- Learn: Neural network basics. Watch the 3Blue1Brown series on neural networks.

Week 2 — Launch fatembed seed 1

- Task: Launch the run: `uv run python -m research.tinystories.train --arm fatembed --vocab 32768 --steps 3000 --target-core 559000 --seed 1` on your Colab account.
- Deliverable: The run reaches 3000 steps. The checkpoint and the run JSON record land in `runs/`.
- Verify: Checkpoints save to Google Drive every 1000 steps.
- Learn: Embeddings and next-token prediction. Explain the difference between ple and fatembed.

Week 3 — Export the ple model

- Task: Run the export on the completed ple checkpoint: `uv run python -m research.tinystories.export ple-cleandeploy-s0 --tokenizer data/tinystories/vocab-32768/tokenizer.json --allow-unverified-tokenizer`.
- Deliverable: The artifacts folder with model.bin, tokenizer.json, and the golden files.
- Verify: The export succeeds and the model.bin size is near 14.9 MB.
- Learn: The transformer block. Explain where the output head sits.

Week 4 — Sampling quality

- Task: Run sample.py on the trained models. Record 20 generations per arm.
- Deliverable: The sampling quality table.
- Verify: The text is coherent at the level of the published demo.
- Learn: Sampling and generation. Explain temperature and greedy decoding.

Week 5 — On-chip profiling

- Task: Profile the runtime on the board (with E1). Measure the stage times: head, attn, ple, ffn, input.
- Deliverable: The per-stage ms/token table vs the published profile.
- Verify: Your profile sums near the total ms/token.
- Learn: Attention math. Write the QKV attention as numpy code.

Week 6 — Serial prompting integration

- Task: Integrate the serial prompting feature into the firmware (with E2).
- Deliverable: A firmware build where the user types a prompt over serial.
- Verify: The board generates text from a typed prompt.
- Learn: Fine-tuning and PTQ. Explain why the model needs no retraining for prompting.

Week 7 — End-to-end demo firmware

- Task: Build the final demo firmware: prompt in, story on the screen.
- Deliverable: The working demo build in the club repo.
- Verify: The demo runs for 10 minutes without a crash.
- Learn: Prepare your teach-back on tokenization and embeddings.

Week 8 — Final demo recording

- Task: Record the final demo: text generation on the board with the tok/s display.
- Deliverable: The demo video in the club Drive.
- Verify: R1 reviews the video before the showcase.
- Learn: Write the "what I learned" page.

Learning path

1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

Resources

- Repo: <https://github.com/slvDev/esp32-ai>
- runtime/host_verify: <https://github.com/slvDev/esp32-ai/tree/main/runtime>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi